

# Advanced vibration suppression algorithms in adaptive optics systems

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Vibration suppression in astronomical adaptive optics (AO) systems has gathered great attention in the context of next-generation instrumentation for current telescopes and future Extremely Large Telescopes. Laser tomographic AO systems require natural guide stars to measure the low-order modes such as tip-tilt (TT) and TT-anisoplanatism. To increase the sky coverage, the guide stars are often faint, thus requiring lower temporal sampling frequencies to work on a more favorable signal-to-noise regime. Such sampling frequencies can be of the order of, or even lower than, the range of frequencies where vibrations are likely to appear. Ideally, vibrations affecting the low-order modes could be corrected at the higher laser loop frame rate using an upsampling procedure. This paper compares the most relevant solutions proposed hitherto to a novel multirate algorithm using the linear-quadratic-Gaussian (LQG) approach capable of upsampling the correction to further reduce the impact of vibrations. Results from numerical Monte Carlo simulations span a large range of parameters from pure sinusoids to relatively broad peak vibrations, covering the likely-to-be signals in a realistic AO system. The improvement is shown at sampling frequencies from 20 to 800 Hz, including below the vibration itself, in the example of 29.5 Hz on a Thirty Meter Telescope-like scenario. The multirate LQG ensures the least residual for both faint and bright stars for all the peak widths considered based on telemetry from the Keck Observatory. © 2012 Optical Society of America

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## 1. VIBRATIONS IN ADAPTIVE OPTICS SYSTEMS

By inducing random fluctuations in incoming stellar wavefronts, atmospheric turbulence is a major factor limiting the performance of ground-based telescopes [1,2]. This limitation can be overcome with an adaptive optics (AO) system. A basic AO system uses a deformable mirror (DM) to counter the distorted wavefronts. The shape of the DM is adjusted in real time based on measurements of the corrected wavefronts by a wavefront sensor (WFS). A more advanced AO system might include more than one WFS and/or more than one DM.

In addition to the atmosphere, observatories themselves generate their own disturbances, often in the form of vibrations. These originate from a myriad of different sources: structural vibrations of the telescope and prefocus mirrors due to wind buffeting, rigid body resonances, motor rotation, power ripples, cryogenic pumps, cooling fans, and hydraulic bearing systems amongst others. Vibrations can directly affect the propagation of the science light, often by causing additional image jitter (tip-tilt). Vibrations can also perturb the operation of the AO system by corrupting the WFS path only, i.e., non-common-path disturbances. These cause the AO system to correct for disturbances that actually have no effect on the science light.

Vibrations affect the majority of the AO systems currently in operation, despite sustained efforts to either remove their sources or isolate them from the light paths. See [3–5] for the ESO's Very Large Telescope, Gemini, and Keck Observatories, respectively, and, more recently, the Large Binocular Telescope [6]. Vibrations often occur at frequencies near or beyond the AO bandwidth and, thus, may be poorly corrected.

This results in excessive residual image jitter, which broadens the core of the point spread function and reduces the Strehl ratio.

The problem with vibrations is even more critical for the proposed next generation of 20–40 m Extremely Large Telescopes (ELTs), because the residual jitter must be kept to within a small fraction of the diffraction limit, a mere few milliarcseconds (mas) rms; the same applies to coronagraph-based planet imagers on the current generation of 8–10 m telescopes, because the efficiency of most coronagraphs significantly drops if even the slightest image jitter is present.

### A. Existing Vibration Suppression Algorithms

Dedicated controllers specifically able to cancel out vibrations have been proposed in the context of the Very Large Telescope's SPHERE and the Gemini Planet Imager instruments [7,8]. These are based on the linear-quadratic-Gaussian (LQG) framework under the implicit assumption that the AO sampling frequency is considerably higher than the frequencies of the vibrations. While this assumption is generally true for planet finder types of systems, which always use a bright guide star and thus typically run at 1–2 kHz sampling frequencies, it will not hold in many cases on more general purpose AO systems, such as the ones envisioned at first light for ELTs. This is so because though these systems use laser guide stars (LGSs), low-order modes such as tip-tilt (TT) and TT-anisoplanatism must be sensed on natural guide stars. These must be allowed to be faint in order to achieve a reasonable sky coverage. For example, in NFIRAOS, the proposed first-light AO instrument for the Thirty Meter Telescope (TMT), the median AO frame rate for tip-tilt control is 90 Hz [9]. This is to

be compared with the natural frequencies of the expected vibrations, typically in the 30–200 Hz range (see, e.g., [4]). When the AO sampling frequency is low compared to the vibration frequencies, the LQG-based methods previously proposed fail to provide an adequate level of rejection.

Another interesting approach is the adaptive cancellation algorithm, initially proposed in [10] for stellar interferometry. This algorithm uses a local oscillator whose frequency, phase, and amplitude are adjusted on-the-fly using a recursive least-squares algorithm. It has been adapted to AO by V  ran et al. [11] under the name AVCA for *adaptive vibration cancellation algorithm*. While the AVCA can potentially correct for vibrations whose frequency is of the order of, or even higher than, the AO sampling frequency by relying on an internal model to upsample the correction, it intrinsically assumes that the vibrations are single-frequency. Considering that vibrations arise from a myriad of sources in the observatory, this assumption is likely to be unrealistic; real vibrations most probably are not single-frequency signals. In Section 4, it is shown that this is indeed the case on the Keck telescopes, based on actual accelerometer measurements.

## B. Multirate Algorithms

The LQG-based multirate approach (MR-LQG) that is proposed here for vibration suppression does not have the shortcomings of the two previous methods; it can suppress vibrations with frequencies comparable to or even lower than the AO sampling frequency, and it can handle relatively broad vibration peaks, as well as disturbances originating from the atmospheric turbulence. The MR-LQG method follows the initial work of [12], where a double-frame-rate model is proposed to allow for an upsampled correction. It relies on a continuous-time representation of the disturbances, from which the coarse- and fine-grain discrete-time models can be built.

This paper is organized as follows. In Section 2 the single-rate LQG controller derivation is presented as a preamble to the multirate LQG controller of Section 3. In Section 4 the performance of the novel multirate LQG is compared with that of the single-rate LQG and that of the AVCA. Section 5 concludes this paper.

## 2. VIBRATION SUPPRESSION ALGORITHMS

We have the standard second-order differential equation

$$\ddot{\theta}(t) + 2\xi\omega_0\dot{\theta}(t) + \omega_0^2\theta(t) = \omega_0^2\nu(t), \quad (1)$$

where  $\theta$  is the vibration signal,  $\xi$  is a damping coefficient (defining the width and overshoot of the vibration peak),  $\omega_0$  is the natural oscillatory frequency in rad/s, and  $\nu$  is a continuous excitation signal.  $\dot{\theta}(t)$  and  $\ddot{\theta}(t)$  represent respectively the first and second order time derivatives of  $\theta(t)$ .

The impulse response of Eq. (1) for  $0 \leq \xi < 1$  is an oscillatory response of the form

$$\theta(t) = \frac{\omega_0 e^{-\xi\omega_0 t}}{\sqrt{1-\xi^2}} \sin\left(2\pi\sqrt{1-\xi^2}\omega_0 t\right). \quad (2)$$

The response envelope defines a constant-width single-frequency signal for  $\xi = 0$  or a damped signal converging exponentially toward zero for  $\xi > 0$ . When  $\nu(t)$  is a zero-mean

Gaussian white noise, the signal  $\theta(t)$  presents a power spectral density marked by a resonance centered at  $f_{\text{vib}} = \frac{1}{2\pi}\omega_0\sqrt{1-\xi^2}$  Hz with FWHM  $\Delta f \approx \sqrt{12}\xi f_{\text{vib}}$  in hertz.

A convenient manner to represent linear differential equations of order  $n$  is to adopt a state-space representation with  $n$  differential equations of the first order [13].

With this representation, Eq. (1) admits the equivalent continuous-time state-space representation (time dependence omitted):

$$\begin{cases} \dot{x}^{\text{vib}} &= A_{\text{vib}}x^{\text{vib}} + \Gamma_{\text{vib}}\nu, \\ \theta &= C_{\text{vib}}x^{\text{vib}}, \end{cases} \quad (3)$$

with

$$\begin{pmatrix} \ddot{\eta} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} -2\xi\omega_0 & -\omega_0^2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \dot{\eta} \\ \eta \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \nu, \quad (4a)$$

$$\theta = \begin{pmatrix} 0 & \omega_0^2 \end{pmatrix} \begin{pmatrix} \dot{\eta} \\ \eta \end{pmatrix}, \quad (4b)$$

where the state  $x^{\text{vib}} \in \mathbb{R}^{2 \times 1}$  involves the dummy variable  $\eta$  and its temporal derivatives. In this particular model,  $\theta(t) = \omega_0^2\eta$ , though, in general,  $\theta$  is obtained as a linear combination of the states. Matrices  $A_{\text{vib}}$ ,  $\Gamma_{\text{vib}}$ , and  $C_{\text{vib}}$  are standard components of state-space models. They are fully defined by Eq. (4).

AO systems are meant to correct for the wavefront disturbances by applying commands  $u_k$ ,  $k \geq 0$  to a DM to counter the wavefront deformation (see Fig. 2). The command set  $u_k \in \mathcal{U}$  that minimizes the variance of the residual wavefront is known to maximize the Strehl ratio [1].

The minimum-variance (MV) LQG controller is designed in two steps: (1) a linear model of the disturbance and its temporal evolution is needed to build the LQ controller in state feedback form, i.e., the real-time controls are obtained from the state as

$$u_k = -\mathcal{K}_\infty x_k^{\text{vib}}, \quad (5)$$

with  $\mathcal{K}_\infty$  the optimal gain, and (2) when the disturbances are measured via a sensor with Gaussian noise, the state in Eq. (5) can be replaced by the estimated state through a Kalman filter. The latter relies on a model of the measurement to produce conditional mean estimates of the state.

Since only disturbance dynamics is considered (the DM is assumed infinitely fast), the optimal control is a simple projection of the disturbance onto the deformable mirrors; hence,

$$u_k = N^\dagger \bar{\theta}_{k+1}, \quad (6)$$

where  $N \in \mathbb{R}^{n_a \times n_\phi}$  is the DM influence matrix of size number of DM commands  $n_a \times$  number of disturbance modes  $n_\phi$ , the superscript  $\dagger$  stands for pseudoinverse, and

$$\bar{\theta}_{k+1} \triangleq \frac{1}{T} \int_{kT}^{(k+1)T} \theta(t) dt \quad (7)$$

is the average vibration amplitude throughout the interval  $t \in [kT, (k+1)T]$ .  $N^\dagger$  is the optimal gain that would be otherwise obtained by solving for the associated control Riccati equation,  $\mathcal{K}_\infty \triangleq -N^\dagger$ .

For a step-by-step derivation of the LQ solution, refer to [14]. How to obtain the required models for the Kalman filter is covered next.

### A. LQG + vib: Single-Rate LQG Controller

Petit *et al.* [15] used a discretized version of the vibration in Eq. (1) that led to the  $\mathcal{Z}$ -domain model

$$H(z) = \frac{\varepsilon_k}{1 - 2e^{-\xi\omega_0^2 T_s} \cos(\omega_0 T_s \sqrt{1 - \xi^2}) z^{-1} + e^{-2\xi\omega_0^2 T_s} z^{-2}}, \quad (8)$$

where  $T_s$  is the AO sampling interval and  $\varepsilon_k$  is the process excitation noise, considered zero-mean, Gaussian-distributed, and temporally white with  $\Sigma_\varepsilon$  covariance matrix.  $T_s$  is kept throughout the document as the AO sampling interval.

The correspondence with a second-order autoregressive formulation is straightforward (using standard  $\mathcal{Z}$ -transform properties [16]):

$$\theta_{k+1} = \alpha_1 \theta_k + \alpha_2 \theta_{k-1} + \varepsilon_k, \quad (9a)$$

$$\alpha_1 = 2e^{-\xi\omega_0^2 T_s} \cos(\omega_0 T_s \sqrt{1 - \xi^2}), \quad (9b)$$

$$\alpha_2 = e^{-2\xi\omega_0^2 T_s}. \quad (9c)$$

In AO, these vibrations are measured via a wavefront sensor. It integrates light emanating from a distant guide star over periods of  $T_s$  s. Assuming a one-frame delay due to camera readout and pixel processing, a model of measurements  $y_k$  corrupted by noise  $w_k$  is

$$y_k = D \underbrace{\frac{1}{T_s} \int_{(k-1)T_s}^{kT_s} \theta(t) dt}_{\hat{\theta}_k} - D N u_{k-1} + w_k, \quad (10)$$

where  $D \in \mathbb{R}^{n_m \times n_\phi}$  is the phase-to-measurements matrix of size number of measurements  $n_m \times$  number of disturbance modes  $n_\phi$ ,  $w_k$  is a discrete-time zero-mean Gaussian distributed white noise of known covariance matrix  $\Sigma_w$ , and

$$u_{k-1} \triangleq u(t), \quad t \in [(k-1)T_s, kT_s], \quad (11)$$

the DM commands applied by a *zero-order hold*.

To be able to use Eq. (9) in modeling the measurement of Eq. (10), let us assume for now that the discretized values of the disturbance signal are reasonably good approximations of the average values over a sampling interval; hence,  $\theta_k = \bar{\theta}_k$ . This constraint will be relaxed in Section 3 in order to increase the accuracy of the model and as a preparatory step for the multirate vibration cancellation algorithm.

From Eq. (9), the discrete-time state-space representation for a standard closed-loop system becomes  $\forall k \geq 0$ ,

$$\begin{cases} x_{k+1}^{\text{vib}} &= \mathcal{A}_{\text{vib}} x_k^{\text{vib}} + \mathcal{B}_{\text{vib}} u_k^{\text{vib}} + \Gamma_{\text{vib}} \varepsilon_k \\ y_k &= \mathcal{C}_{\text{vib}} x_k^{\text{vib}} + w_k \end{cases} \quad (12)$$

(calligraphic notation used for discrete-time matrices instead of roman notation), with the fully populated model

$$\underbrace{\begin{pmatrix} \theta_{k+2} \\ \theta_{k+1} \\ u_k \end{pmatrix}}_{x_{k+1}^{\text{vib}}} = \underbrace{\begin{pmatrix} \alpha_1 & \alpha_2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\mathcal{A}_{\text{vib}}} \underbrace{\begin{pmatrix} \theta_{k+1} \\ \theta_k \\ u_{k-1} \end{pmatrix}}_{x_k^{\text{vib}}} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathcal{B}_{\text{vib}}} u_k^{\text{vib}} + \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}_{\Gamma_{\text{vib}}} \varepsilon_k, \quad (13a)$$

$$y_k = \underbrace{D(0 \quad 1 \quad -N)}_{\mathcal{C}_{\text{vib}}} x_k^{\text{vib}} + w_k. \quad (13b)$$

Note that a greater integer measurement delay is easily modeled by including in the state time-shifted versions of the discretized phase and commands. Fractional delays that result in asynchronous operation can be coped with as well [17].

With this model, the optimal estimator can be readily derived [13], as is shown next.

### B. Final Controller Design

The Kalman filter is obtained from an estimation Riccati equation. Since the metric of interest in AO is the long exposure integration of light on the science instruments, such that  $T_i \gg T_s$ , the asymptotic solution can be used. The optimal gains are computed off-line as follows:

$$\mathcal{L}_\infty = \mathcal{A}_{\text{vib}} \mathcal{H}_\infty = \mathcal{A}_{\text{vib}} \Sigma_\infty \mathcal{C}_{\text{vib}}^\top (\mathcal{C}_{\text{vib}} \Sigma_\infty \mathcal{C}_{\text{vib}}^\top + \Sigma_w)^{-1}, \quad (14)$$

where  $\Sigma_\infty$  is the solution of the algebraic Riccati equation

$$\begin{aligned} \Sigma_\infty &= \mathcal{A}_{\text{vib}} \Sigma_\infty \mathcal{A}_{\text{vib}}^\top + \Sigma_\varepsilon \\ &\quad - \mathcal{A}_{\text{vib}} \Sigma_\infty \mathcal{C}_{\text{vib}}^\top (\mathcal{C}_{\text{vib}} \Sigma_\infty \mathcal{C}_{\text{vib}}^\top + \Sigma_w)^{-1} \mathcal{C}_{\text{vib}} \Sigma_\infty \mathcal{A}_{\text{vib}}^\top. \end{aligned} \quad (15)$$

The only unknown in the previous equation is the covariance of the discrete-time excitation signal  $\varepsilon_k$ . It is found from

$$\Sigma_\varepsilon = \Sigma_\theta \left( 1 - \alpha_1^2 \alpha_2^2 - \frac{2\alpha_1^2 \alpha_2}{1 - \alpha_2} \right), \quad (16)$$

which is the solution to the discrete-time Lyapunov equation

$$\Sigma_\theta = \mathcal{A}_{\text{vib}} \Sigma_\theta \mathcal{A}_{\text{vib}}^\top + \Sigma_\varepsilon, \quad (17)$$

relating the total disturbance covariance matrix to the noise covariance that produces it. A robust method to identify  $\omega_0$ ,  $\xi$ , and  $\Sigma_\theta$  on the fly from closed-loop telemetry is reported in [7].

The controller is applied in real time by computing, at iteration  $k$ ,

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + \mathcal{H}_\infty (y_k - \mathcal{C}_{\text{vib}} \hat{x}_{k|k-1}), \quad (18a)$$

$$\hat{x}_{k+1|k} = \mathcal{A}_{\text{vib}} \hat{x}_{k|k} + \mathcal{B}_{\text{vib}} u_k^{\text{vib}}, \quad (18b)$$

$$u_k = [1 \quad 0 \quad 0] \hat{x}_{k|k} = N^\top \hat{\theta}_{k+1}, \quad (18c)$$

where the hat represents the conditional mean estimation of the state. The commands  $u_k$  use an orthogonal projection of the estimated vibration state  $\theta_{k+1}$  (first component of  $x_k^{\text{vib}}$ ) onto the DM influence-functions-spanned vectorial space [Eq. (6)].

### 3. TOWARD A MULTIRATE LQG CONTROLLER

As discussed previously, the AO WFSs integrate starlight over a period of  $T_s$  s, where  $T_s$  must be long enough to reach sufficient signal-to-noise ratio on the WFS. This sets the WFS frame rate, and thus the AO loop frame rate, to  $f_s \triangleq 1/T_s$ , with the DMs updated at that rate via a zero-order hold. The underlying idea of the multirate LQG is that while the WFS frame rate cannot be reduced due to the limited number of photons available, the DM update rate could be increased to better track the disturbance.

When the WFS frame rate is less than about 10× the central frequency of the vibration, the accuracy of the measurement process is compromised. Figure 1 shows the ratio between the variance of a harmonic signal after averaging (i.e., as seen by the WFS) and the variance of the underlying continuous representation of this signal. For sampling frequencies 100 times the vibration frequency, 99.9% of the signal can be potentially recovered. For a factor of 10, this decreases to roughly 97%.

Therefore, the shortcoming of using the approximation  $\theta_k = \bar{\theta}_k$  is that unless the WFS sampling frequency is very high, the time averaging of the wavefront in the WFS is not being properly modeled.

The alternative pursued here is to start from a continuous-time model of the vibration and discretize it in order to obtain accurate average values of the measured wavefront, as output by the WFS.

Starting from Eq. (4), the discrete-time model that has  $\theta_k \triangleq \theta(kT_s)$ ,  $\forall k \geq 0$  (referred to in the following as “exactly discretized” to indicate it has the same values at the sampling instants  $kT_s$ ,  $\forall k \geq 0$ ) is [18]

$$\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}_{k+1} = e^{A_{\text{vib}} T_s} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}_k + \varepsilon_k^{\text{vib}}, \quad (19a)$$

$$\theta_k = \begin{pmatrix} 0 & \omega_0^2 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}_k, \quad (19b)$$

where  $\eta_1$  and  $\eta_2$  are general discrete-time states. The discrete-time white noise sequence is given by  $\varepsilon_k^{\text{vib}} = \int_0^{T_s} e^{A_{\text{vib}}(T_s-s)} \nu(kT_s + s) ds$ , whereas the covariance of discrete-time noise  $\varepsilon_k^{\text{vib}}$  is obtained from  $\Sigma_{\varepsilon^{\text{vib}}} = \int_0^T e^{A_{\text{vib}} T_s} \Sigma_{\nu} e^{A_{\text{vib}}^T T_s}$ , with  $\Sigma_{\nu}$  the spectrum of  $\nu(t)$ . For a complete derivation, refer to [18], Sections 4.4 and 4.5.

The average of the vibration over a sampling interval can be obtained from the discretized state [14] as

$$\begin{aligned} \bar{\theta}_{k+1} &= \frac{1}{T_s} \int_{kT_s}^{(k+1)T_s} \theta(t) dt \\ &= \frac{1}{T_s} C_{\text{vib}} (e^{A_{\text{vib}} T_s} - I) A_{\text{vib}}^{-1} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}_k \\ &= \Xi_{\text{vib}} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}_k. \end{aligned} \quad (20)$$

Under the hypothesis of a loop delay equal to exactly one frame (see Section 2.A), an augmented state-space model can thus be determined:

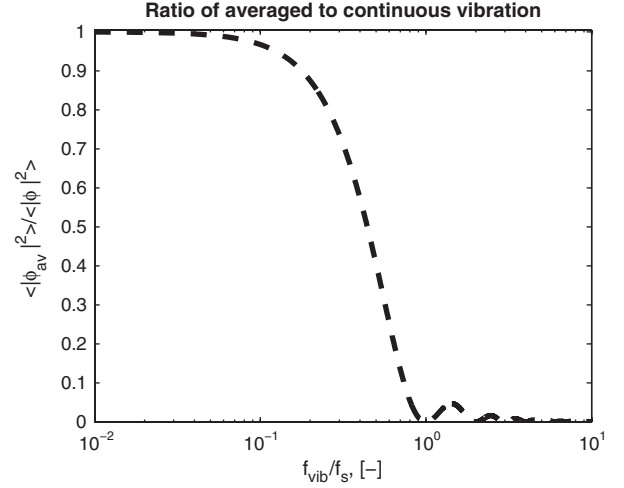


Fig. 1. Variance ratio of a vibration averaged over a  $1/f_s$  width window to its original continuous representation, given by a sinc<sup>2</sup> function. For  $f_s = 10f_{\text{vib}}$  the ratio is 0.968, whereas for  $f_s = 100f_{\text{vib}}$  the ratio is 0.999.

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \bar{\theta}_{k+1} \\ \bar{\theta}_k \\ u_k \end{pmatrix} = \begin{pmatrix} e^{A_{\text{vib}} T_s} & 0 & 0 & 0 \\ \Xi_{\text{vib}} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \\ \bar{\theta}_k \\ u_{k-1} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_k + \varepsilon_k, \quad (21a)$$

$$y_k = D \begin{bmatrix} 0 & 0 & 0 & 1 & -N \end{bmatrix} x_k^{\text{vib}} + w_k, \quad (21b)$$

where the zeros represent matrices of the proper sizes. The noise  $\varepsilon_k$  now has a more complex expression due to the propagation of state noise  $\varepsilon_k^{\text{vib}}$  onto the average disturbance  $\bar{\theta}$ . A thorough derivation of the noise expression is provided in Appendix B of [14].

Having decoupled the discretized signals at the sampling instants  $t = kT_s$ ,  $\forall k \geq 0$  from the averaged signals in this model, it is now straightforward to use the multirate algorithm to produce a correction signal at a higher frame rate.

#### A. MR-LQG: Multirate LQG Controller

The developments in this section build on the approach of [12]. First a fine-grain model of the disturbance is needed, from which the upsampled commands can be computed. This can be easily obtained by discretizing the continuous-time state-space models at the fine-grain sampling rate.

The fine-grain vibration model (using the subscript  $f$ ) is

$$\begin{cases} x_{i+1}^{\text{vib}} &= \mathcal{A}_{\text{vib},f} x_i^{\text{vib}} + \epsilon_i \\ \bar{\theta}_i &= \Xi_{\text{vib},f} x_i^{\text{vib}} \end{cases}, \quad (22)$$

with  $\mathcal{A}_{\text{vib},f} = e^{A_{\text{vib}} T_f}$  and  $\Xi_{\text{vib},f} = \frac{1}{T_f} C_{\text{vib}} (e^{A_{\text{vib}} T_f} - I) A_{\text{vib}}^{-1}$ , discretized at  $T_f < T_s$ , a higher frame rate than the nominal coarse-grain model of Eq. (19). The integer ratio of the coarse and fine sampling times is  $n = T_s/T_f$ .

The following coarse-grain state-space model based on Eq. (22) can be constructed as

$$\begin{cases} x_{k+1}^{\text{vib}} &= (\mathcal{A}_{\text{vib},f})^n x_k^{\text{vib}} + \varepsilon_k^{\text{vib,MR}} \\ \theta_k &= \Xi_{\text{vib}} x_k^{\text{vib}} \end{cases}, \quad (23)$$

where  $(\mathcal{A}_{\text{vib},f})^n \triangleq \mathcal{A}_{\text{vib}}$ . For the average vibration over the sampling interval  $T_s$  from the  $n$  fine-grain  $\bar{\theta}_{\frac{i}{n}}$  contributions, one gets

$$\bar{\theta}_{k+1} = \frac{\Xi_{\text{vib},f}}{n} \sum_{i=0}^{n-1} (\mathcal{A}_{\text{vib},f})^i \quad (24)$$

$$= \frac{\Xi_{\text{vib},f}}{n} \sum_{i=0}^{n-1} (e^{\mathcal{A}_{\text{vib}} T_f})^i. \quad (25)$$

Taking limits leads to

$$\lim_{n \rightarrow +\infty} \frac{\Xi_{\text{vib},f}}{n} \sum_{i=0}^{n-1} (e^{\mathcal{A}_{\text{vib}} T_f})^i = \frac{1}{T_s} \int_0^{T_s} e^{\mathcal{A}_{\text{vib}} t} dt \quad (26)$$

$$= \frac{1}{T_s} C_{\text{vib}} (e^{\mathcal{A}_{\text{vib}} T_s} - I) \mathcal{A}_{\text{vib}}^{-1} \quad (27)$$

$$= \Xi_{\text{vib}}, \quad (28)$$

which is indeed an expected result (the average over  $T_s$  is the average of averages over  $T_f$ ; note that  $\Xi_{\text{vib},f} \rightarrow 1$  when  $n \rightarrow \infty$ , since  $T_f \rightarrow 0$ ). This slight modification with respect to [12] is found to have better numerical behavior.

The noise covariance matrices are now slightly different for the multirate ( $n > 1$ ) case [12]:

$$\Sigma_{\varepsilon^{\text{vib}}}^{\text{MR}} = \sum_{i=0}^{n-1} (\mathcal{A}_{\text{vib},f})^i \Sigma_{\varepsilon} (\mathcal{A}_{\text{vib},f}^T)^i, \quad (29a)$$

$$\begin{aligned} \Sigma_w^{\text{MR}} &= \Sigma_w + \frac{1}{n^2} D C_{\text{vib}} \sum_{i=0}^{n-1} \\ &\times \left[ \left( I + \sum_{j=0}^i (\mathcal{A}_{\text{vib},f})^j \right) \Sigma_{\varepsilon} \left( I + \sum_{j=0}^i (\mathcal{A}_{\text{vib},f})^j \right)^T \right] C_{\text{vib}}^T D^T, \end{aligned} \quad (29b)$$

$$\Sigma_{\varepsilon^{\text{vib},w}}^{\text{MR}} = \frac{1}{n} D C_{\text{vib}} \sum_{i=1}^n \left[ (\mathcal{A}_{\text{vib},f})^i \Sigma_{\varepsilon} \left( I + \sum_{j=0}^{i-1} (\mathcal{A}_{\text{vib},f})^j \right)^T \right] C_{\text{vib}}^T D^T. \quad (29c)$$

For the single-rate case ( $n = 1$ ), they reduce to

$$\Sigma_{\varepsilon^{\text{vib}}}^{\text{MR}} = \Sigma_{\varepsilon^{\text{vib}}}, \quad (30a)$$

$$\Sigma_w^{\text{MR}} = \Sigma_w, \quad (30b)$$

$$\Sigma_{\varepsilon^{\text{vib},w}}^{\text{MR}} = 0. \quad (30c)$$

The correlation between the state noise  $\varepsilon_k$  and the measurement noise  $w_k$  is due to the effect of the  $n - 1$  command updates over each  $T_s$ -wide interval of WFS measurement.

The augmented state-space model is now constructed as before, noting that the coarse-grain state-space model is identical to Eq. (21a). In other words, the multirate model based on the fine-grain state-space model is identical to the coarse-grain state-space running at the WFS's sampling frequency. This is particularly attractive from the real-time computation point of view. The only difference is with the measurements, where now the effects of the controls on the residual wavefront need to be accurately removed due to intersample actuation to produce a commonly called "pseudo-open-loop measurement"  $y_k^{\text{ol}} = D \bar{\phi}_k^{\text{tur}} = D(\bar{\phi}_k^{\text{res}} + \bar{\phi}_k^{\text{cor}})$ . Therefore the model is

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \theta_{k+1} \\ \theta_k \\ u_k \end{pmatrix} = \underbrace{\begin{pmatrix} [e^{\mathcal{A}_{\text{vib}} T}] & 0 & 0 & 0 \\ [\Xi_{\text{vib}}] & 0 & 0 & 0 \\ [0] & 1 & 0 & 0 \\ [0] & 0 & 0 & 0 \end{pmatrix}}_{\mathcal{A}^{\text{MR}}} \begin{pmatrix} \eta_1 \\ \eta_2 \\ \theta_k \\ u_{k-1} \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathcal{B}^{\text{MR}}} u_k + \varepsilon_k^{\text{MR}}, \quad (31a)$$

$$y_k^{\text{ol}} = D \underbrace{([0 \ 0] \ 0 \ 1 \ -N)}_{\mathcal{C}^{\text{MR}}} x_k + w_k^{\text{MR}}. \quad (31b)$$

The operations to be executed in real time are now [compare to Eq. (18)]

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + \mathcal{H}_{\infty} (y_k^{\text{ol}} - \mathcal{C}^{\text{MR}} \hat{x}_{k|k-1}), \quad (32a)$$

$$\hat{x}_{k+1|k} = \mathcal{A}^{\text{MR}} \hat{x}_{k|k} + \mathcal{B}^{\text{MR}} u_k^{\text{MR}}, \quad (32b)$$

$$u_{k+i/n} = N^{\dagger} [\Xi_{\text{vib},f} (\mathcal{A}_{\text{vib},f})^i] \hat{x}_{k|k}^{\eta}, \quad (32c)$$

with the pseudo-open-loop measurement given by

$$y_k^{\text{ol}} = y_k + D N \frac{1}{n} \sum_{i=0}^{n-1} u_{k-1+i/n} \quad (33)$$

and the estimated state given by the conditional mean at instant  $k$ , knowing all past measurement and command history up to instant  $k$ :

$$\hat{x}_{k|k}^{\eta} \triangleq \begin{bmatrix} [c] c \hat{\eta}_1 \\ \hat{\eta}_2 \end{bmatrix}_{k|k}. \quad (34)$$

For the multirate case, the off-line estimation Riccati equation [Eq. (15)] is

$$\begin{aligned} \Sigma_{\infty}^{\text{MR}} &= \mathcal{A}_{\text{vib}} \Sigma_{\infty}^{\text{MR}} \mathcal{A}_{\text{vib}}^T + \Sigma_{\varepsilon} - (\Sigma_{\varepsilon}^{\text{MR}} + \mathcal{A}_{\text{vib}} \Sigma_{\infty}^{\text{MR}} C_{\text{vib}}^T) \\ &\times (C_{\text{vib}} \Sigma_{\infty}^{\text{MR}} C_{\text{vib}}^T + \Sigma_w)^{-1} (C_{\text{vib}} \Sigma_{\infty}^{\text{MR}} \mathcal{A}_{\text{vib}}^T + \Sigma_{\varepsilon}^{\text{MR}}), \end{aligned} \quad (35)$$

to account for the correlation of  $\varepsilon_k$  and  $w_k$ , since the WFS measures the multiple command updates throughout the integration interval. Equation (14) is reused to obtain the Kalman filter gains.

### B. Complete Methods: Atmospheric Model Stack with Vibration Add-Ons

For an AO system used to correct for atmospheric turbulence, vibrations or, for that matter, any other known disturbance can be incorporated as an add-on to the model of the atmospheric turbulence. This being said, an augmented model is built, where the disturbances are modeled as independent noise-filtered signals mixed up in the measurement equation, which sums up both the vibrations and the atmospheric disturbances.

For clarity, this model includes only one single mode (e.g., either tip or tilt) but can be generalized to any number of modes. It is detailed in the following equations:

$$x_k \triangleq \begin{pmatrix} x_k^{\text{tur}} \\ x_k^{\text{vib},1} \\ \vdots \\ x_k^{\text{vib},n} \\ \phi_{k-1}^{\text{total}} \\ u_{k-1} \end{pmatrix}, \quad A \triangleq \begin{pmatrix} A_{\text{tur}} & 0 & 0 & 0 & 0 & 0 \\ 0 & A_{\text{vib},1} & 0 & 0 & 0 & 0 \\ 0 & \cdots & \ddots & \vdots & \vdots & 0 \\ 0 & \cdots & \cdots & A_{\text{vib},n} & 0 & 0 \\ C_{\text{tur}} & C_{\text{vib},1} & \cdots & C_{\text{vib},n} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (36a)$$

$$B \triangleq \begin{pmatrix} B_{\text{tur}} \\ B_{\text{vib},1} \\ \vdots \\ B_{\text{vib},n} \\ 0 \\ 1 \end{pmatrix}, \quad \Gamma \triangleq \begin{pmatrix} \Gamma_{\text{tur}} & 0 & 0 & 0 & 0 & 0 \\ 0 & \Gamma_{\text{vib},1} & 0 & 0 & 0 & 0 \\ 0 & \cdots & \ddots & \vdots & \vdots & 0 \\ 0 & \cdots & \cdots & \Gamma_{\text{vib},n} & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{pmatrix}, \quad (36b)$$

$$C \triangleq D(0 \quad \cdots \quad 0 \quad 1 \quad -N), \quad (36c)$$

$$C^{\text{MR}} \triangleq D(0 \quad \cdots \quad 0 \quad 1 \quad 0), \quad (36d)$$

with

$$x_k^{\text{vib}} \triangleq \begin{pmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \theta_k \end{pmatrix}_k, \quad A_{\text{vib}} \triangleq \begin{pmatrix} e^{A_{\text{vib}} T_s} & 0 \\ \Xi_v & 0 \end{pmatrix}, \quad (37a)$$

$$B_{\text{vib}} \triangleq 0, \quad \Gamma_{\text{vib}} \triangleq I, \quad (37b)$$

$$C_{\text{vib}} \triangleq ([0 \quad 0] \quad 1), \quad (37c)$$

and

$$x_k^{\text{tur}} \triangleq \begin{pmatrix} x_{k+1}^{\text{tur}} \\ \phi_{k+1}^{\text{tur}} \end{pmatrix}, \quad A_{\text{tur}} \triangleq \begin{pmatrix} e^{T_s A_{\text{tur}}} & 0 \\ \Xi_{\text{tur}} & 0 \end{pmatrix}, \quad (38a)$$

$$B_{\text{tur}} \triangleq 0, \quad \Gamma_{\text{tur}} \triangleq I, \quad (38b)$$

$$C_{\text{tur}} \triangleq ([0 \quad 0] \quad 1), \quad (38c)$$

where  $I$  is the identity matrix of the proper size.

### C. Hybrid Single-Rate Atmospheric Multirate Vibration LQG Controller

Using the model stack above, the controller can be split into two parts: one for correcting for atmospheric turbulence and another for correcting for vibrations. In order to illustrate the benefit of the multirate algorithm for vibration suppression, the higher correction frame rate was implemented only for the vibration part. In a real implementation, all the corrections can be implemented at a higher frame rate, although the benefit for the mostly low-frequency atmospheric turbulence is likely to be less spectacular.

In most cases, the DM dynamics can be neglected, which leads to

$$u_k = K_{\infty} \bar{\phi}_{k+1|k}^{\text{total}} = N^{\dagger} \left( \underbrace{\hat{x}_{k+1|k}^{\eta}}_{\text{vibration}} + \underbrace{\hat{\phi}_{k+1|k}^{\text{tur}}}_{\text{turbulence}} \right) \quad (39)$$

for the single-rate case and

$$u_{k+i/n} = K_{\infty,i}^{\text{vib}} \hat{x}_{k+i/n|k}^{\eta} + K_{\infty}^{\text{tur}} \hat{\phi}_{k+1|k}^{\text{tur}} = N^{\dagger} \left( [\Xi_{\text{vib},f} \quad (A_{\text{vib}})^i] \quad 1 \right) \begin{pmatrix} \hat{x}_{k|k}^{\eta} \\ \hat{\phi}_{k+1|k}^{\text{tur}} \end{pmatrix} \quad (40)$$

for the multirate case. The first term of  $K_{\infty,i}^{\text{vib}}$  produces averaged correction signals over  $T_f = T_s/n$  intervals (fine-grain) that interpolate the vibration. The multiplication by the matrix  $(A_{\text{vib},f})^i$  predicts the vibration at step  $k+i/n$ , whereas the matrix  $\Xi_{\text{vib},f}$  averages it out over  $T_f$ . Finally, the second part produces the turbulence correction using the coarse-grain frame rate. A useful feature of this algorithm is that the Kalman filter runs at the WFS's frame rate. The multirate up-samples the correction based on  $T_s$  estimates of the state, thus only requiring a  $[1 \times 2]$  matrix multiplication per update, per vibration, and per mode. These are the  $n$  values of  $[\Xi_{\text{vib},f} (A_{\text{vib}})^i]$ ,  $i \in 0 \cdots n$ , which can be precomputed.

## 4. SAMPLE NUMERICAL RESULTS

The AO systems to be installed on ELTs will use several laser guide stars (LGSs) to probe the three-dimensional atmospheric volume. However, TT and TT-anisoplanatism cannot be measured with LGSs and will be sensed using natural guide stars (NGSs). Most often, the only NGSs available will be faint, which will lead to large integration times  $T_s$ . For example, in NFRAOS [19,20], the first-light multiconjugate AO facility for TMT, the NGS WFS frame rate is expected to be in the 20–800 Hz range, with a median of 90 Hz [9], with NGSs as faint as magnitude 22 (H-band). Other tomographic AO systems combining NGSs and LGSs such as the E-ELT MAORY and ATLAS modules [21] will operate in roughly the same parameter range. At such low frame rates, vibration rejection will be a big challenge.

To assess the benefit of the multirate LQG algorithm in this context, a numerical simulation was used, following the block diagram of Fig. 2. Continuous variables are considered to be sampled with at least a  $10\times$  higher frame rate than any other frame rates in the system. Only vibrations affecting image

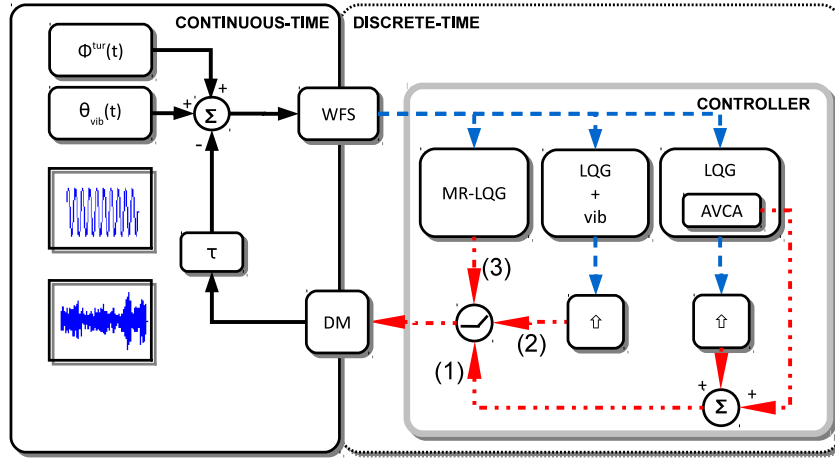


Fig. 2. (Color online) Closed-loop control architecture of a standard AO system. Solid lines for continuous-time signals and dashed lines for discrete-time signals (dash-dotted red for  $f_{\max} = 800$  Hz and dashed blue for  $f_s \leq f_{\max}$ ). The upsampling blocks (upward pointing arrows) are zero-order-hold blocks. The two windows below  $\theta_{\text{vib}}$  represent a pure sinusoid and Gaussian white noise filtered by a second-order model (damped vibrations), respectively—see Fig. 3. Both these disturbances are used as input in the simulations. A constant servo lag of  $\tau = 1$  ms is used to represent real-time computer delays in the loop.

jitter are modeled, which only involves the subset of TT modes. As is common in many AO systems, TT is corrected via an off-load steering mirror and an actual DM. For the sample results, an abstraction is made: TT is corrected with an infinitely fast, unbounded stroke mirror.

Three controller configurations are compared against each other with

(i) LQG + AVCA: A single-rate LQG controller with no internal vibration suppression embedded coupled to the AVCA algorithm—the optimal Kalman filter uses the model stack of Eq. (36) with no vibrations in it, with  $\mathcal{C}$  in Eq. (36c) and commands from Eq. (39).

(ii) LQG + vib: A single-rate LQG controller with vibration suppression embedded running at the NGS WFS's sampling frequency  $f_s = 1/T_s$ .

CASE (ii-a): This controller used either the AR2 vibration model proposed in [15] rewritten here in Section 2.A, Eq. (13).

CASE (ii-b): The controller uses the exactly discretized model of Eq. (37). It corresponds to the filter using the full model Eq. (36), with  $\mathcal{C}$  in Eq. (36c) and command given by Eq. (39).

(iii) MR-LQG: A multirate LQG controller with vibration suppression embedded running at the NGS WFS's sampling frequency  $f_s$  and outputting commands at the maximum frame rate of  $f_{\max} = 800$  Hz, as explained in Section 3.A. The Kalman filter is derived from the model of Eq. (36), with  $\mathcal{C}^{\text{MR}}$  in Eq. (36d) and the multirate command in Eq. (40).

The simulation parameters are summarized in Table 1. A single vibration peak is used, centered at  $f_{\text{vib}} = 29.5$  Hz, which is a commonly measured vibration in telescopes. This disturbance is generated by pumps and fans driven by moderately loaded alternating-current four-pole induction motors.

The FWHM of the vibrations used are motivated by those extracted from telemetry at the Keck telescope [22] Figure 3 shows the PSD of TT acquired with accelerometers attached to the AO system parts. Four major peaks were identified at [1.85; 9.8; 29.4; 225] Hz with FWHM ranging from  $27.4 \times 10^{-3}$  to 1.37 Hz carrying different levels of power.

For the numerical simulation, the full range of these parameters is explored. However, though the vibrations' band and peak width can be extrapolated to real-case ELTs, the total disturbance energy is unknown. Therefore, the same rms value for the turbulence + wind-shake is used as vibration. The results are shown in relative units rather than absolute units to make them convertible to other applications.

### A. Purely Sinusoidal Vibrations

Figure 4 shows normalized residuals for all the controllers' variations—CASE (i) through CASE (iii)—when the vibration is a pure sinusoid at  $f_{\text{vib}} = 29.5$  Hz.

Table 1. Parameters Used in the End-to-End Monte Carlo Numerical Simulations<sup>a</sup>

|              |                    |                                                                                                       |
|--------------|--------------------|-------------------------------------------------------------------------------------------------------|
| Disturbances | Atmospheric        | $\sigma_{\text{TT}} = 16.58$ mas rms<br>$L_0 = 30$ m<br>$r_0 @ 0.5 \mu\text{m} = 0.15$ m              |
|              | Wind shake         | $\sigma_{\text{WS}} = 7.5$ mas rms                                                                    |
|              | Vibration          | $\sigma_{\text{vib}} = 18.8$ mas rms<br>$f_{\text{vib}} = 29.5$ Hz<br>FWHM = {1.37, 0.274, 0.0274} Hz |
| WFS          | Type               | Modal                                                                                                 |
|              | Frame rates        | {20, 50, 100, 200, 400, 800} Hz                                                                       |
|              | Read + dect. noise | {1.1, 11} mas rms                                                                                     |
| Controllers  | LQG                | w/AVCA—CASE (i)                                                                                       |
|              | LQG                | w/disc. vib. model—CASE (ii)                                                                          |
|              | Multirate LQG      | w/disc. vib. model—CASE (iii)                                                                         |
|              | Disturbance models | Order 2                                                                                               |
|              | Servo lag          | 1 ms                                                                                                  |
| Correctors   | Type               | Modal                                                                                                 |
|              | Frame rate         | 800 Hz                                                                                                |
|              | DM                 | Infinite bandwidth                                                                                    |
| Simulation   | Duration           | $10 \times 45$ s                                                                                      |
|              | No. modes          | 1 (tip or tilt)                                                                                       |
|              | Wavelength         | $\lambda_{\text{WFS}} = 1.6 \mu\text{m}$                                                              |

<sup>a</sup>The WFS noise values are equivalent to those of an H-band star with magnitudes 13 and 20, sampled at 800 Hz and 20 Hz, respectively, which covers a broad range of NGS stars used for wavefront sensing in a TMT-NFIRAOS-like configuration.

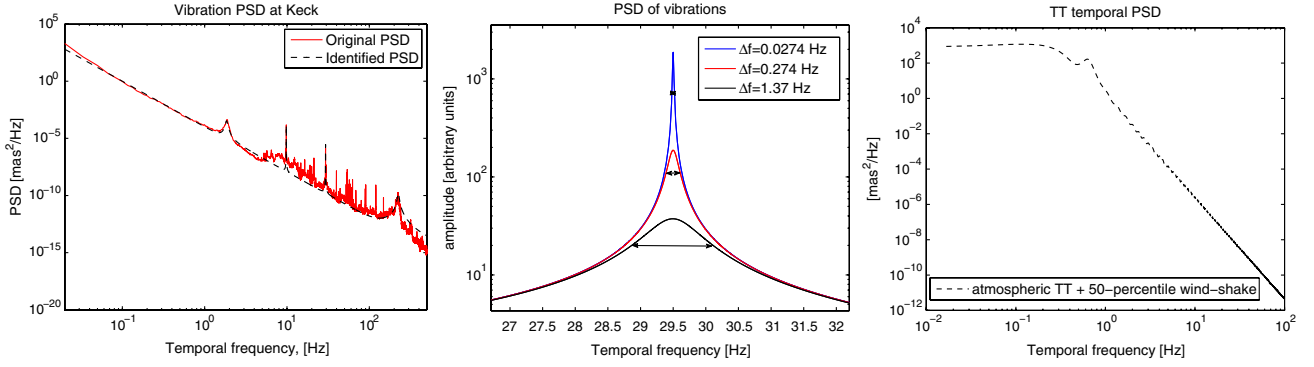


Fig. 3. (Color online) Left: Vibration PSD at Keck with identified models of the second order at frequencies 1.85, 9.8, 29.4, and 225 Hz. Center: Vibrations chosen to illustrate the numerical results. Model parameters are indicated in the legend of the figure. Right: Atmospheric TT+ wind-shake disturbance power spectral density.

Several comments are in order. First, the performance is roughly the same for all three configurations at  $f_s = 800$  Hz. The increased performance of the MR-LQG is clear with respect to the LQG + vib for all  $f_s < f_{\max}$ . The latter tends to 1 (i.e., the uncorrected vibration disturbance) when  $f_s$  approaches the lower bound (here, 20 Hz), whereas the MR-LQG ensures an attenuation factor of 5 in rms units. The LQG + AVCA for  $f_s < 2f_{\text{vib}}$  breaks down, since the recursive least-squares algorithm tracking vibration parameters no longer converges. For  $f_s > 2f_{\text{vib}}$ , the AVCA is no better than the MR-LQG, since the residuals are alike.

It should be emphasized that the performance obtained with all the controllers assumes that an accurate model of the vibration exists and is fed to the LQG controller's design and/or to the AVCA. This is why the MR-LQG is able to track vibrations at  $f_s < 2f_{\text{vib}}$ , even though the vibration is being almost completely averaged out in the WFS measurements. This presupposes an accurate auxiliary means to identify vibration parameters, for example, accelerometer measurements such as those shown in Fig. 3.

To understand why both the single-rate LQG using an AR2 and an exactly discretized vibration model achieve the same

performance with a pure-sinusoid vibration, note that the rejection transfer functions in Fig. 5 show an identical rejection at  $f_{\text{vib}}$ . If the vibration falls right in that rejection notch, the residuals are indeed the same. However, when the vibration presents some peak width, then the side lobes of the LQG rejection transfer function with an AR2 vibration model [Eq. (13)] tend to amplify considerably more than the LQG with the exactly discretized vibration model [Eq. (21)]. Hence, the exactly discretized model is preferable even for single-rate applications.

## B. Damped Vibrations

Figure 6 shows the performance for vibration disturbances obtained from the model of Eq. (4) with an FWHM varying from 0.0274 to 1.37 Hz, as shown in Fig. 3. The top and bottom panels explore two rms measurement noise values, corresponding to relatively bright and faint natural guide stars (see Table 1).

The superiority of the MR-LQG is clear in these examples. It is always better than the single-rate controllers, and with regard to the AVCA, the MR-LQG at low sampling frequencies outperforms the latter by factors of up to 5 (in rms reduction).

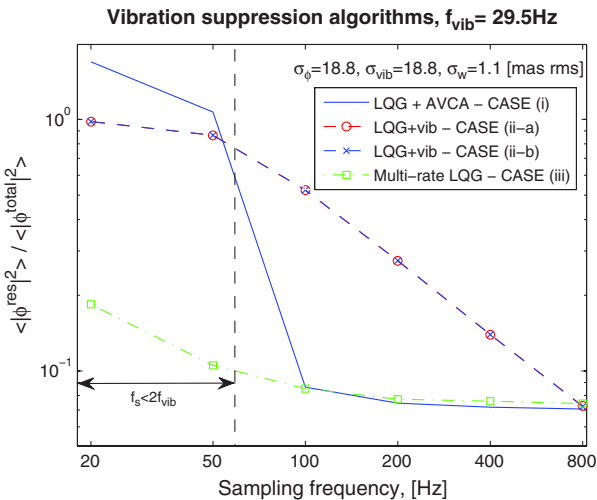


Fig. 4. (Color online) Comparison of vibration suppression algorithms when the vibration is a constant-amplitude, constant-frequency monochromatic sinusoid. Solid line: AVCA. Dashed line: Single-rate LQG. Dash-dotted line: Multirate LQG.

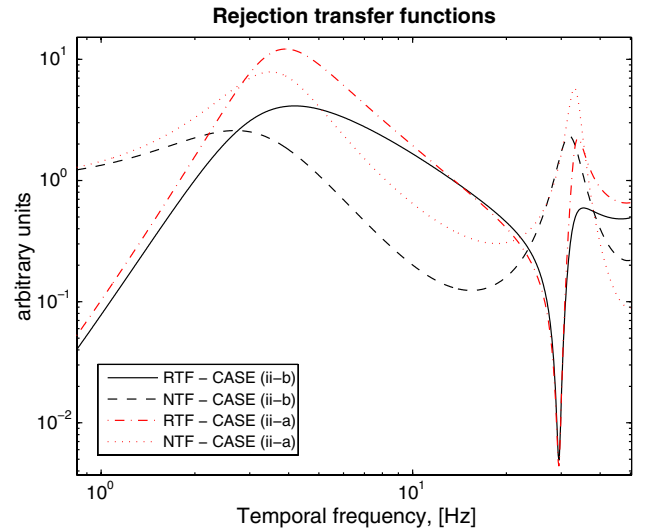


Fig. 5. (Color online) Signal and noise transfer functions at  $f_s = 100$  Hz for the LQG controllers using an exactly discretized vibration model (black) or an AR2 model (red).

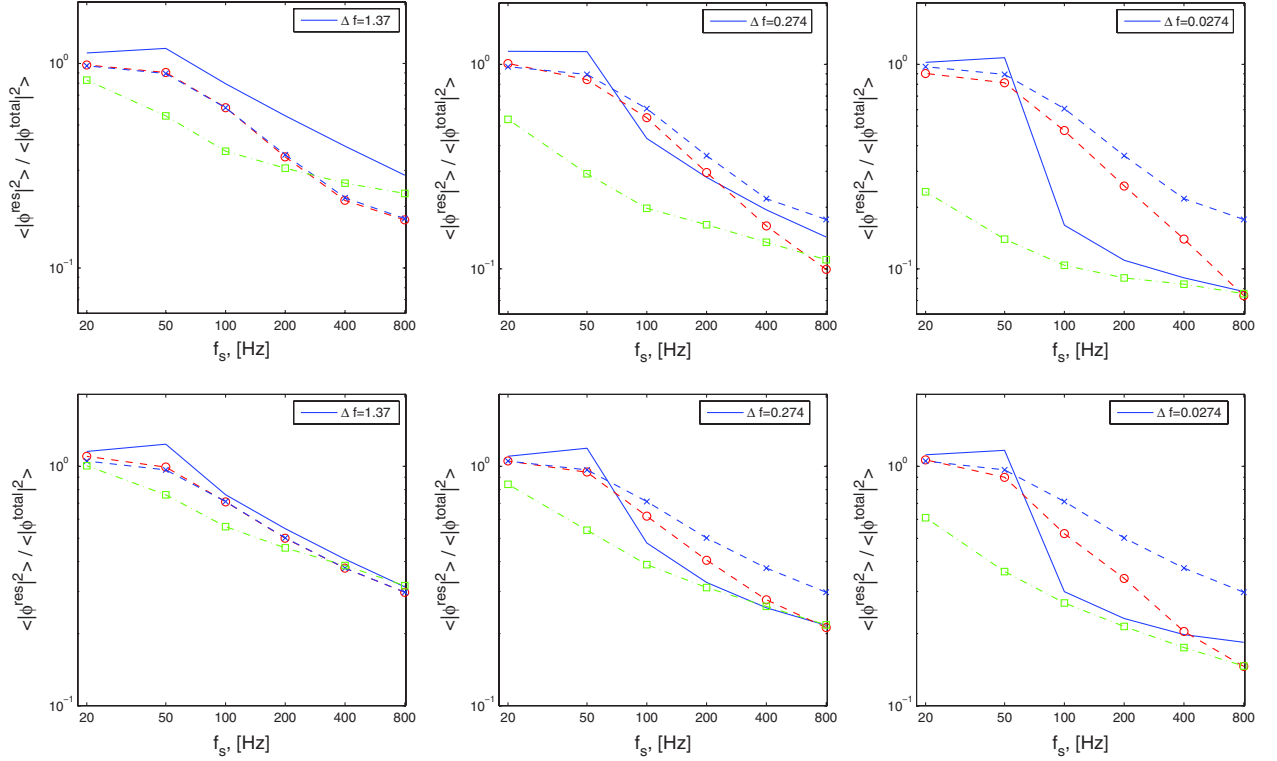


Fig. 6. (Color online) Performance comparison of the new multirate algorithm with the single-rate one using a continuous phase model, a single rate using a discrete phase model [CASES (ii-a) and (ii-b)], and the AVCA (see full-color key in Fig. 4). Upper panel:  $\sigma_w = 1.1$  mas rms. Bottom panel:  $\sigma_w = 1.1$  mas rms, relative to two working points with different star magnitudes (see Table 1).

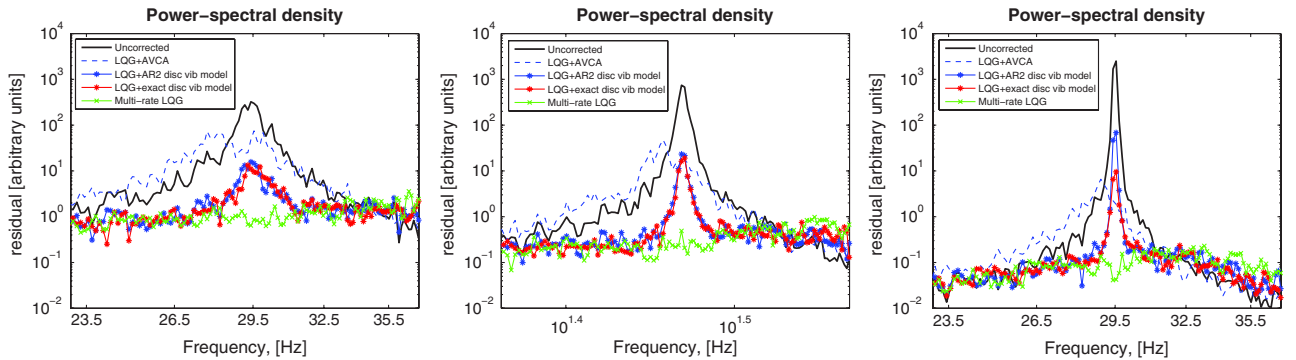


Fig. 7. (Color online) Power-spectral density plots zoomed around  $f_{\text{vib}} = 29.5$  Hz of the uncorrected disturbance and the residual obtained with three vibration suppression strategies for  $f_s = 100$  Hz and (left to right)  $\Delta f = \{1.37, \dots, 0.0274\}$ . The multirate approach strongly smears the vibration peak, whereas the AVCA leaves a broadband residual. The single-rate LQG + Arkansas2 model [CASE (ii-a)] is not well suited for vibration suppression at this frame rate, since its vibration rejection is ineffective (see Fig. 1).

The leftmost column shows that when the vibration peak is relatively large (FWHM = 1.37 Hz) and the sampling frequency is below  $\approx 50$  Hz, no actual estimate is done, averaging out the vibration being the best the controllers are able to do. The ratios thus tend naturally to 1. This is particularly true, in general, for high noise values, as is shown in the bottom panel.

Compared to the LQG + AVCA, every LQG + vib controller, and in particular the MR-LQG, can better adapt its internal model to the vibration's. This model is not bounded to pure or quasi-pure vibrations.

Figure 7 depicts the uncorrected and residual PSD for the three FWHM considered (Table 1) at  $f_s = 100$  Hz, for the lowest of the noise levels used. The MR-LQG is the only controller

capable of completely smearing the vibration peak, guaranteeing the flattest residual. The single-rate controllers leave a considerable residual peak, whereas the AVCA tends to broaden the residual around  $f = f_{\text{vib}}$ , a consequence of its attempt to continuously track the central frequency, without ever perfectly matching it as a consequence of the input not being monochromatic (as its model expects and predicts).

## 5. CONCLUSION

The Monte Carlo numerical simulation for vibrations on the tip-tilt modes compares four different vibration suppression algorithms. It is shown that the new multirate LQG method is

superior to single-rate controllers and even to the AVCA. The latter is very well suited for narrowband, slow evolving vibrations with domain of application  $\Delta f \lesssim 0.03$  Hz and  $f_s > 2f_{\text{vib}}$ , cases in which the MR-LQG performs roughly equally. For all the remaining cases,  $\Delta f \gtrsim 0.03$  Hz and  $f_s < 2f_{\text{vib}}$ , the multi-rate LQG outperforms both the single-rate controllers, on account of the interpolation of the disturbance during the sampling interval, and also the AVCA, since the latter is not fully able to track the vibration variations in amplitude and central frequency.

Given the likelihood of finding moderately large vibration peaks on a real AO system, multirate vibration suppression ensures better overall rejection at all sampling rates and for all vibratory peak widths.

The multirate controller can be seen as an upsampling procedure based on *a priori* knowledge of the temporal evolution of the disturbances. This is particularly useful for optimal data blending from natural and laser guide stars, where the LGS WFS runs at a frame rate that is greater than or equal to that of the NGS WFS. This multimode scenario, along with the case of multiple vibration peaks, will be treated in a forthcoming paper.

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